

**DESIGN AND DEVELOPMENT OF AN EXTREMELY WIDEBAND
MINIATURIZED MONOCONE ANTENNA WITH HIGH GAIN
ATTRIBUTES**

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DEDICATION

This thesis is dedicated to my late grandma, Gauri. And my parents Krishna and Kantibhai for instilling values, morals and hard work in my life.

PREVIEW

Thoroughly conscious ignorance is the prelude to every real advance in science

– James C. Maxwell

ABSTRACT OF THE THESIS

Design and Development of an Extremely Wideband Miniaturized
Monocone Antenna with High Gain Attributes

by

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The wideband miniaturized antennas are readily used in the industry for sensing, communication, direction finding and other ultrawideband applications. Miniaturized antennas with smaller form factor are more attractive for their flexibility and mobility. Often, such miniaturized antennas suffer from low gain and small bandwidth. This thesis will try to find solutions offering high gain and large bandwidth for miniaturized antennas covering 2 to 20 GHz. For wideband miniaturization, different reactive loading techniques will be explored. A generalized approach to obtain such miniaturized for any operating frequency will be provided in this thesis, such that the technique can be utilized to cover any frequency bands with certain bandwidth from scaled versions of the antenna. Fundamental limits of miniaturization will be studied to design the antenna to provide both wideband and certain level of miniaturization. Protective structures such as dielectric radomes will be investigated without affecting the original antenna performance. Different gain enhancement techniques including metal cavity backed and dielectric lens-based designs will be discussed. The thesis will also focus on fabricating the antenna with 3D printed technology for metal and dielectric structures. A hybrid structure will be designed using both metal cavity backed design and radome to provide protect and gain enhancement. A wideband antenna source will be designed, fabricated using 3D printed technology and tested for antenna performance. A wideband gain enhancement technique will be designed, 3D printed and experimentally verified. The antenna will be especially measured for impedance matching and radiation pattern characteristics verification. The thesis will also demonstrate practical applications of the wideband miniaturized antenna in Multiple Input Multiple Output (MIMO) and Unmanned Aerial Vehicles (UAVs).

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PREVIEW

CHAPTER 1

INTRODUCTION

1.1 MOTIVATION

Commercialization of fifth generation (5G) and development of sixth generation (6G) communication systems has garnered wide interest in various antenna designs to achieve high data rate and realize intelligent wireless systems. Additionally, many of these antennas are placed in a space-constrained environment making compact antennas highly desirable. Often, multiple antennas are used to accommodate different application bands. Designing a single compact or miniaturized antenna that covers multiple bandwidths can significantly save space. These wideband miniaturized antennas are often required to cover Wi-Fi (2.4 and 5 GHz), UWB (3.1 to 10.8 GHz), satellite communication bands in X-band (8.2-12.4 GHz) and Ku-band (12-18 GHz) [1], [2]. Such antennas can be used in current and future MIMO arrangements to achieve high data throughput and channel reliability. These wideband miniaturized antennas can also be used in unmanned aerial vehicles (UAVs) to establish different types of communication at various operating frequencies.

Designing a wideband miniaturized antenna requires to tackle two different aspects: one designing a wideband antenna source and second to employ a miniaturization technique without compromising the wide bandwidth properties of the original antenna source. This thesis will delve into some of the wideband miniaturization techniques with a greater focus on designing the antenna operating from 2-20 GHz. Depending on the type of application, these antennas may require radome structures to protect them from environmental hazards without compromising the antenna performance. The integration with a UAV platform may require the antennas to operate at much higher gain. Hence, gain enhancement techniques including cavity and dielectric lens designs are explored to meet this requirement.

In this thesis, the main objectives are: (i) design of a miniaturized wideband antenna operating from 2-20 GHz, (ii) design of a dielectric radome to provide protection from environmental hazard without affecting the antenna performance, (iii) explore the hemispherical cavity, dielectric lenses and a new hybrid cavity-lens designs to enhance the antenna gain throughout the frequency of operation, (iv) study the performance of the proposed antenna in a 4x4 MIMO arrangement, and (v) integrate the proposed antenna onto a simple UAV platform to observe its performance.

1.2 LITERATURE REVIEWS

In this section we discuss about different state of the art works related to the wideband miniaturized antenna, gain enhancement techniques using metal-backed cavity and dielectric lenses, MIMO and UAVs.

1.2.1 Wideband Antennas

Different types of wideband antennas reported in literature are shown in Figure 1.1(a-g). In [2], a planar compact super wideband antenna covering 1.44 – 18.8 GHz is reported by embedding a fractal-complementary slot into an asymmetrical ground plane of an egg-shaped monopole antenna. The fractal-complementary slot aids in impedance matching towards the lower operating frequency thereby increasing the operating bandwidth. Semi-elliptical slots are used on an uneven ground plane to shift the frequency from 1.74 to 1.44 GHz and achieve antenna gain of up to 7 dBi at higher frequencies. In [3], three-dimensional (3-D)-printed, wideband, compact hemispherical-shaped antenna is reported which consists of a driven strip monopole and capacitively loaded circular parasitic strips that reside on the surfaces of a hemispherical shell. Achieves -10 dB impedance bandwidth of 17.97% with realized gain values of 3.5 dBi. Particle swarm optimization (PSO) algorithm is used to optimize the gap between the parasitic strip to achieve wider bandwidth. A wideband hemispherical helix is reported in [4] with 4.5 turns with constant spacing, built using tapered metallic strips where different tilt angles are used to tune the frequency of operation. The antenna is fed by a coaxial cable with the inner conductor connected to the radiating strip and outer

conductor connected to a ground plane. The antenna reported in [4] achieves -10 dB fractional impedance bandwidth of 50% and a 3 dB axial ratio bandwidth of 24%.

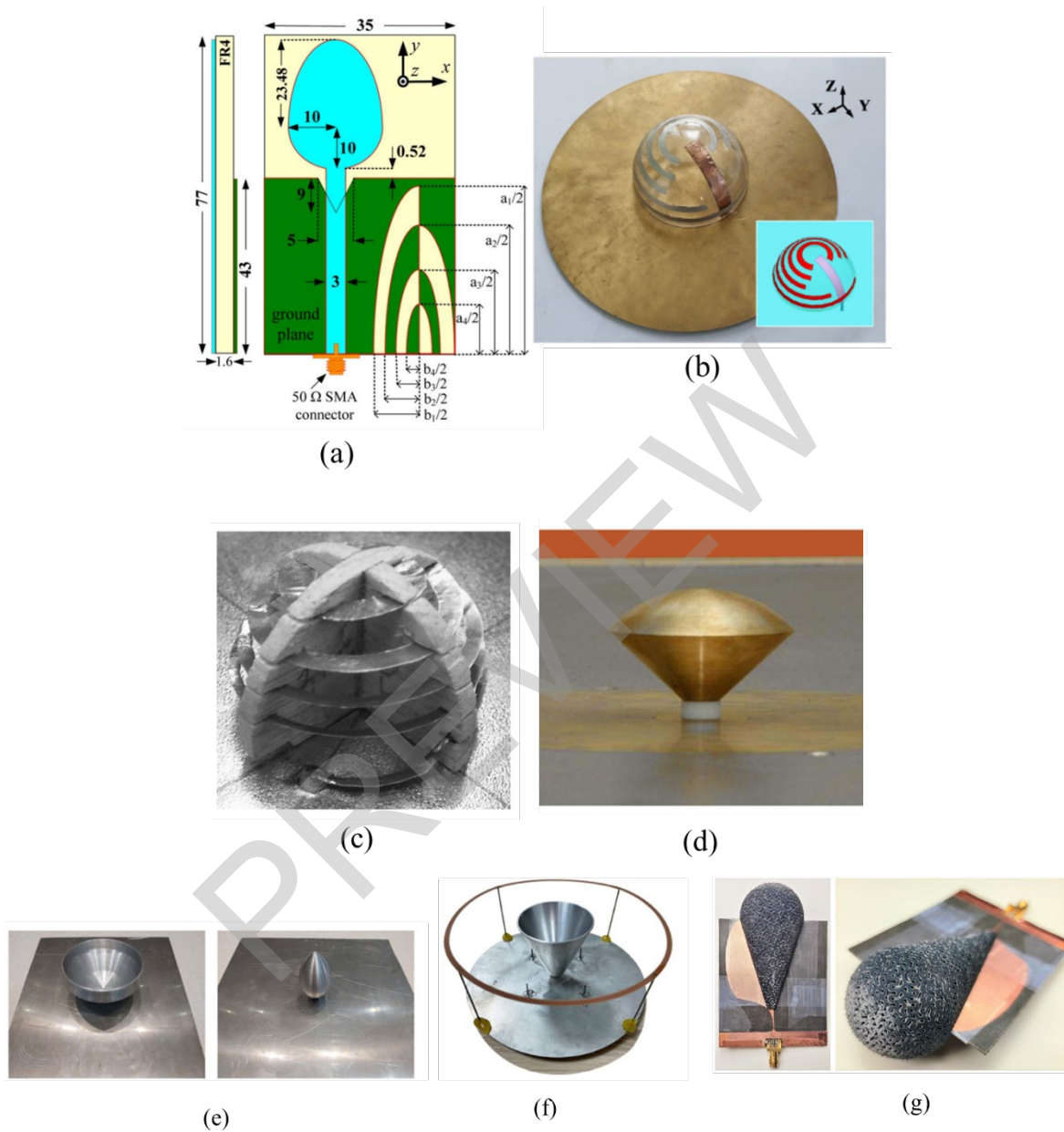


Figure 1.1. Different types of wideband antenna. (a) SWB printed monopole, source: [2] (b) 3D printed wideband hemispherical-shaped driven by a strip monopole, source: [3] (c) Wideband hemispherical helix, source: [4] (d) Wideband monocone to transmit pulse signal, source: [5], (e) Wideband monocone antennas using super curves, source: [6], (f) Monocone with shorted pins, source: [7] and (g) Balanced Antipodal Vivaldi Antennas with hemispherical lens, source: [8].

Similar wideband hemispherical based structures are reported in [9] and [10]. A parasitic wire is integrated in hemispherical helix reported in [9] to enhance the bandwidth. In [10], the advantages of hemispherical helix in terms of axial ratio, power, gain and radiation pattern are discussed. Monocones are 3D antennas used in mobile wireless vehicles and military communication applications due to their wideband, omnidirectional patterns in azimuth plane, compact, and mechanically robustness. Monocones find application in radar sensor networks, various vehicular platforms and reverberation chamber measurements [5], [6], [7], [11], [12], [13]. In [5], a wideband a 40mm long monocone is used to transmit pulse signals. The monocone has a slant angle of 47° designed to provide steady wide impedance and admittance after the resonance. In [6], super curves are utilized to design wideband monocone antennas. The super shaped monocone is created by giving arbitrary super curves to the cone. The lower half is based on cone which is fed with a coaxial connector and the upper consists of arbitrary shape. The shape has length of 16 cm, height of 8.7 cm and a 0.2 cm height difference at the junction of coax cable and the cone. The super shaped monocone antenna is optimized to provide -10 dB impedance bandwidth from 1.4 to 39 GHz with omnidirectional pattern and a gain of 2 to 5 dBi. Furthermore, undesired higher-order mode distorts the radiation patterns. A body of revolution monocone antenna is also designed in [6], where the super curves are applied on the whole structure, creating a continuous curve throughout. The junction between the excitation of the monocone and the ground plane have higher impacts on the impedance bandwidth. The super shape presented has a 10 cm height and 5.3 cm diameter. It has S_{11} lower than -10 dB from 0.8 to 55 GHz with gain ranging from 3 to 6.5 dBi. The pattern becomes more "conical" at higher frequency. The bandwidth is higher than the half super shaped monocone structure. A monocone with shorted pins to the ground improve radiation pattern over a wide bandwidth [5]. Similar, wideband monocone antennas are designed in [11], [12], [13], [14] using shorting pins, modified feeding transition, etc. Modified monocone in [12] included cone and a cylinder at the junction between the ground plane and the lower half of cone. Merging these two curves results in an increased impedance bandwidth. The cone angles higher than 59° provided wider bandwidth and lower antenna profile compared to classical monocone. A monocone is modified to fill the null pattern by

adding rotational current distribution in [14]. A slotted path is added in the reference monocone with a pitch angle of 45° . The impedance matching of 1:12, 160 MHz to 1960 MHz and a nearly circular polarization (CP) is achieved for aircraft mountable communication antenna where CP would provide better communication. In [8] a balanced antipodal Vivaldi antenna is also reported to achieve wide antenna bandwidth over 15.8:1. The radiation features are often improved using dielectric lenses as reported in [8].

1.2.2 Antenna Miniaturization Techniques

Along with wide-bandwidth, physical size of the antenna is also crucial for various applications. Keeping this in mind various miniaturization techniques are reported in literature [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38]. Significant theoretical limits and relation between parameters such as bandwidth, efficiency and miniaturization factors are reported in [15], [16], [17], [18], [19], [20]. Fundamental limits on the physical size of antennas has been heavily researched since mid-1900s. The concept was refined over the years but has been based on the equations from several researchers who proposed the antenna performance limitations through derivations. Miniaturized antennas are also characterized based on these fundamental limitations. The Chu-Harrington limit characterizes a limit where antenna starts to lose its antenna performance when its physical size reduces. L.J. Chu derived limitation for an omnidirectional antenna, where the relationship between gain, quality factor and bandwidth are found in [15]. Chu's sphere defining the miniaturization level is shown in Figure 1.2. Miniaturization level is often characterized by ka , where k is the free space wavenumber given by $2\pi/\lambda$, and a is the radius of the Chu's sphere.

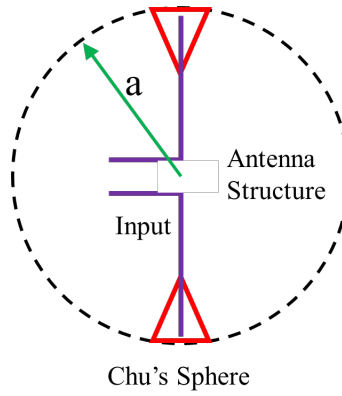


Figure 1.2. Chu's sphere as reported in [15].

Various miniaturization concepts and techniques are reported in [16], [17], [18], [19]. The inverse relationship between quality factor and bandwidth is given in [19] for multiple resonant antennas. It is illustrated that for closely spaced resonances, the S_{11} is less than -10 dB for wider bandwidths. More theoretical study of the fundamental limit for electrically small antennas are given in [20], [21], [22], [23]. Some important miniaturization techniques reported in literature are given in Figure 1.3. In [24], negative refractive index (NRI) metamaterials (MTM) are used to load a monopole antenna, where lumped capacitor and inductors dual topology is implemented to achieve good impedance matching. Four monopole vias are miniaturized by using metamaterial cells encompassing lumped inductor (L) and capacitor (C) to impedance match the antenna without using any external matching network. The antenna has a reduced size of $ka = 0.45$ with a ground plane size of $0.45\lambda \times 0.45\lambda$. However, the design only works over a narrow bandwidth of 53 MHz. Similarly, lumped L and C components are used in [25], [26], [27] to achieve ka values less than 0.5. In [26], a folded dipole loaded with series LC provide miniaturization of up to 75% with a narrow fractional bandwidth of 1.5% is presented. A non-foster matching technique is used in [27], where reactive elements are used with a monopole patch antenna source to provide tunability. It achieves bandwidths from 4 to 8% range. In [29], [30], [31], [32], [33], the use of capacitively loaded loop (CLL) to design miniaturized antennas with $ka < 0.5$ is reported. In [29], a driven semi-circular loop source antenna with a CLL create near field magnetic coupling, providing control over capacitance values which in turn controls the